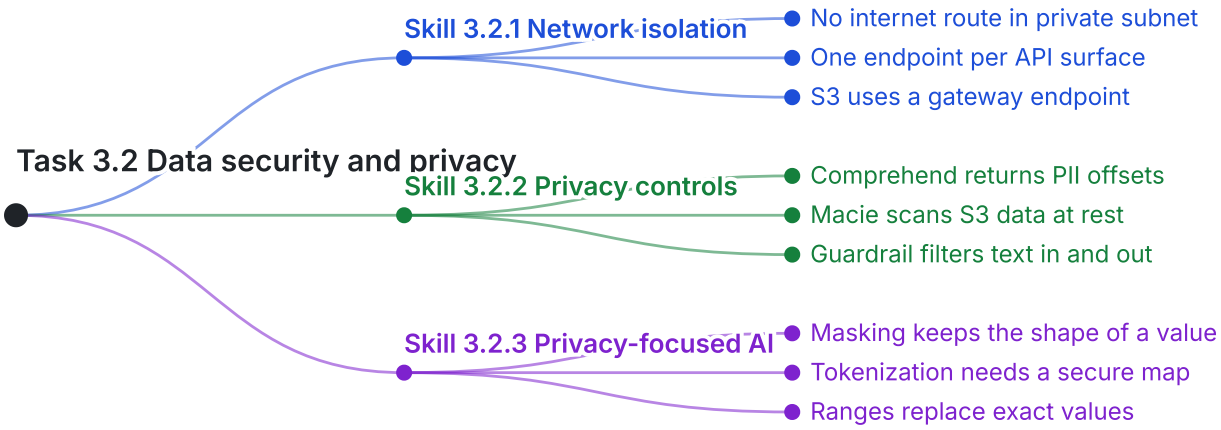
Memory sheet: Task 3.2 Implement data security and privacy controls

Domain 3, weight 20 percent. Read this sheet, then answer the flash questions aloud.



Deciding rules

When the stem says	Think of	Not when
No internet route, or traffic must stay on the AWS network	VPC endpoints for each AWS API the app calls	A NAT gateway route, which still reaches the internet
Data at rest in S3, discovery, or inventory	Amazon Macie	A live chat message, since Macie reads S3 objects only
Live request, before the model call, or exact offsets	Comprehend real-time call, with code that masks each span	A document over 100 KB, which the real-time call cannot take
Block, mask, or a policy on prompts and answers	Bedrock guardrail attached to the model call	Stored S3 files, since a guardrail filters text at inference time
Column, row, or tag based access to a data lake	AWS Lake Formation permissions	A direct S3 path read, which needs IAM or bucket control
Who called, when, or which API call	AWS CloudTrail records	A request for prompt text, since CloudTrail holds no request body
Trend, threshold, or an alarm on access volume	CloudWatch metrics and alarms	Which user read one column, since a metric does not record one request
Deletion after a set number of days or years	S3 Lifecycle expiration action	Data a legal hold still needs, or logs in CloudWatch Logs

Etch lines

1. Each Bedrock API surface needs its own interface endpoint.
2. A gateway endpoint serves S3 and DynamoDB only as a route table target.

3. Hiding one column needs Lake Formation, not an IAM policy.
4. Macie does not scan a live chat message, because it reads S3 objects only.
5. A guardrail mask protects the answer, while the invocation log keeps the original request.
6. With private DNS on, standard hostnames resolve to the endpoint, so code stays unchanged.

Flash questions

- Q: Which endpoint type gives a private subnet S3 access with no NAT gateway? A: A gateway endpoint, added as a route table target in the same Region.
- Q: Which Bedrock interface endpoint serves Converse and InvokeModel calls? A: The bedrock-runtime endpoint. The bedrock endpoint covers control plane actions only.
- Q: A private app calls a model and invokes an agent. What is the minimum endpoint count? A: Two, bedrock-runtime and bedrock-agent-runtime.
- Q: Which condition key shows that a request came through one endpoint? A: aws sourceVpce. It proves the network path, not the caller.
- Q: Which service finds PII in data that is already stored in S3? A: Amazon Macie. It does not scan a live chat message.
- Q: Does the real-time Comprehend PII response contain redacted text? A: No. It returns types, scores, and offsets, so your code applies the change.
- Q: Does masking an answer with a guardrail change what the invocation log keeps? A: No. The log input still holds the original request.
- Q: What does an S3 Lifecycle expiration action do, and what should you check first? A: It deletes objects after a set time. Check for a legal hold first.

Source: Network isolation for FM and agent traffic, Privacy controls for FM interactions, Privacy-focused AI systems masking and anonymization, and The deciding constraint. AWS docs and the study guide sections above, no practice-test text.